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RESEARCH **Open Access**

Experimental demonstration of LED-to-LED visible light communication between a traffic light and a vehicle headlamp

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Abstract

Visible light communication (VLC) technology using light-emitting diodes (LEDs) is currently being discussed to solve the problem of oversaturation in radio frequency (RF) bands. We conducted this study due to a lack of research on LED-to-LED VLC in the context of vehicle communication. One of the VLC technologies, LED-to-LED VLC, is a technology that applies LEDs to both a transmitter and a receiver. In this study, we demonstrate LED-to-LED VLC technology between a vehicle headlamp and a traffic light. We employed LEDs of different colors in the receiver and analyzed the theoretical channel capacity by measuring the 3-dB bandwidth, signal-to-noise ratio (SNR), and channel capacity. The communication performance was confirmed by measuring the bit-error rate (BER) as a function of the data rate. Studies have demonstrated that LED-to-LED VLC is feasible in the field of vehicle communication and is anticipated to be utilized for communication or data collection on the road.

Keywords: LED-to-LED VLC, Optical wireless communication, Visible light communication, Vehicle communication, Signal-to-noise ratio

1 Introduction

The growing use of personal electronic devices and the rapid development of mobile communication technology are increasing the demand for wireless data capacity, which has led to the need for more frequency bandwidth. However, the current RF band is supersaturated, and with the development of communication technology, the mm band is also expected to become supersaturated soon. Therefore, optical wireless communication (OWC) technology using infrared (IR), ultraviolet (UV), and visible light (VL) band has been actively studied as an alternative to this frequency supersaturated state [1–4].

OWC technologies includes various technologies such as visible light communication (VLC), light fidelity (Li-Fi), optical camera communication (OCC), and free space optical communication (FSOC). VLC, one of the OWC technologies, is a communication technology using visible light bands. VLC uses light-emitting diodes (LEDs) or laser diodes (LDs) as transmitters and photodiodes (PDs) or image sensors (ISs) as receivers. The block diagram of a typical VLC system is shown in Fig. 1.

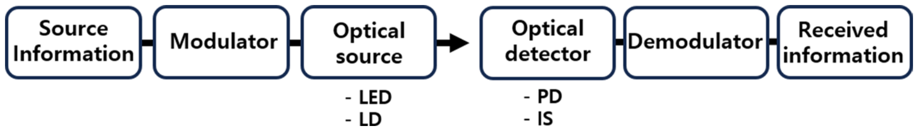

Fig. 1 Block diagram of a typical VLC system

VLC systems are being studied in various fields such as internet of things (IoT), software-defined radio [5], indoor positioning system [6], vehicle-visible light communication [7], and optical-frequency conversion [8]. VLC technology offers several advantages such as wide bandwidth, no electromagnetic interference (EMI), easy implementation of the technology and high security. LEDs are being used in many places because of their longer lifespan and lower power consumption compared to conventional fluorescent lamps.

With LEDs already installed in many places as transmitters, VLC can be implemented economically. However, commercialization of VLC technologies is not fast. We think that one of the main reasons is that VLC requires PDs or ISs for reception. Although LEDs are already installed in many places, we need to install additional hardware, PDs or ISs, as receivers of VLC. Therefore, we studied LED-to-LED VLC, which is a more economical way to implement VLC. LED-to-LED VLC is a VLC technology that uses LEDs as both a transmitter and a receiver.

LEDs consists of PN junctions, like PD, so they can operate as light-emitting devices as well as light-receiving devices. LED functions as a light-emitting device when forward bias is applied. However, if reverse bias or zero bias is applied, LED can function as a light-receiving device like a PD.

LED-to-LED VLC system is a low-cost VLC technology that does not need additional optical receivers such as a PD or a IS. Recently, several LED-to-LED VLC technologies that use LEDs as both transmitters and receivers has been studied [9–11]. Multiple-input multiple-output (MIMO) LED-to-LED VLC with RGB colors has been studied [12]. In [13–15], LED-to-LED VLC systems in both half-duplex and full-duplex modes were studied.

In this work, we utilize LED-to-LED VLC technology in the vehicle communication. With the development of autonomous driving technology of automobiles, we expect that there will be a need for vehicle communication. However, vehicle communication also relies on limited RF bands, with numerous studies focusing on technologies such as advanced driver assistance systems (ADAS) [16] and dedicated short-range communication (DSRC) [16]. Therefore, in recent years, research has been actively conducted to address the issue of RF band supersaturation by applying visible light communication to vehicle communication and adding optical elements such as PD and IS to the receiver. These studies include infrastructure-to-vehicle (I2V) [17], multi-hop vehicle-to-vehicle (V2V) [18], traffic light-to-vehicle VLC [19], V2V VLC [20], and camera-based vehicular VLC [21]. Research has been conducted on the feasibility and application of LED-based VLC systems in vehicle communication [22]. In [22], a study focused on a VLC system designed with an LED transmitter and a photodetector receiver. Additionally, a study has conducted an experiment on a VLC system using vehicle headlamps as the transmitter and a photodetector as the receiver [23]. In [23], experimental results confirmed that communication at 10 kbps is possible over a distance of 20 m using the 4-pulse position modulation (PPM) technique. In contrast, our research explored vehicle communication using an LED-to-LED VLC system that does not require an additional optical detector at the receiver. We thought here that we could study LED-to-LED VLC-based vehicle communication without additional optical elements to gain economic benefits. By applying LED-to-LED VLC technology to vehicle communication, we conducted a communication experiment in which the transmitter is a vehicle headlamp and the receiver is a traffic light. If this technology is used, the headlamp has the advantage of being able to use it for communication purposes that can transmit information as well as for the purpose of lighting. In the future, we expect this technology to be used for data collection or communication on the road. One example of this application is reducing traffic density by collecting road data, including vehicle density and estimated travel time to the destination. Additionally, it is expected that traffic lights, by gathering vehicle-related information, can help improve law enforcement efficiency. In this study, the characteristics of headlamps and traffic lights were identified, and the theoretical channel capacity for transmission was analyzed. In addition, we analyzed the performance of the communication by conducting an LED-to-LED VLC experiment using vehicle headlamps and traffic lights.

2 Methods and experimental setup

In this paper, an LED-to-LED VLC experiment was conducted using vehicle headlamps and traffic lights. In the experiment, the color temperature of the headlamp was 5500 K, and Philips' LED headlamp was used. Also, we used traffic lights of red, yellow, and green colors. The traffic lights were 8×8, a total 64 LEDs. As for the traffic light LED, Yinhui Photoeelectric's LED was used and the size of LEDs is 10 mm $\times$ 10 mm. The transmitter of the experiment was a vehicle headlamp, and the receiver was a traffic light. Figure 2

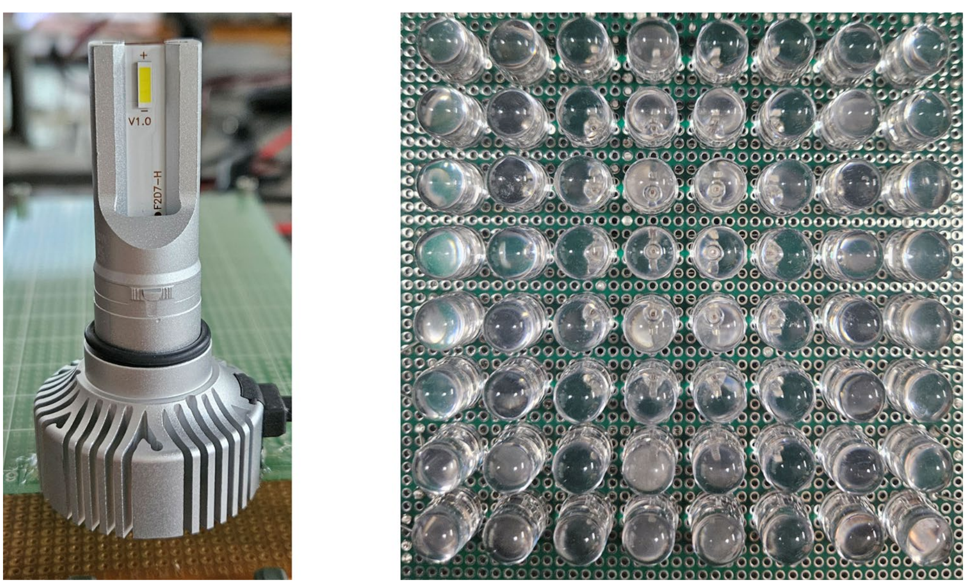

Fig. 2 Photographs of the headlamps and traffic lights used in the experiment

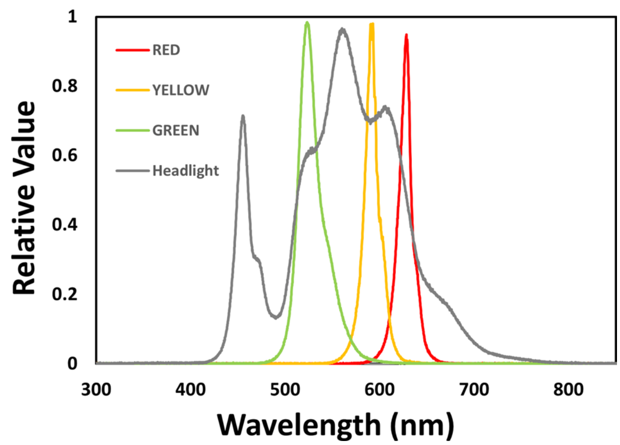

Fig. 3 Optical Spectra **a** headlamps and **b** red, **c** yellow, and **d** green traffic lights

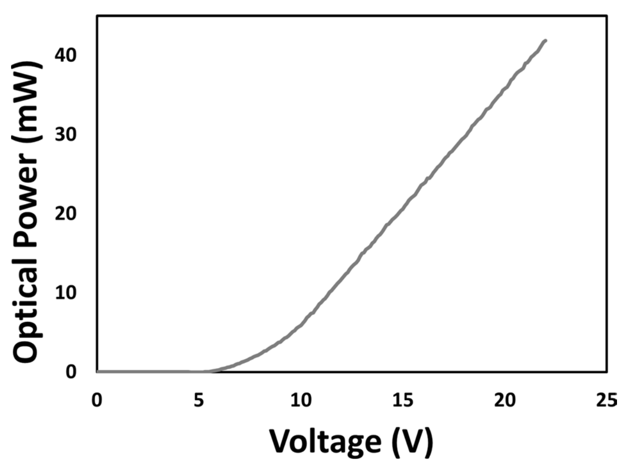

Fig. 4 Optical power of the headlamp according to the input voltage

shows the picture of the vehicle headlamps and traffic lights used in the experiment. The distance setting between the transmitter and receiver is 50 cm. The experiment focused on the scenario where the vehicle is stationary to receive a signal from a traffic light. In addition, the traffic light receiving the signal does so with its light turned off. The transmission and reception in the experiment were conducted using a simplex mode, where information could only be transmitted in one direction.

Meanwhile, it was impossible to make a communication system using a traffic light as a transmitter and a headlamp as a receiver. The reason is that the headlamp is a phosphorescent conversion white LED (pc-white LED). If a pc-white LED is used as a receiver, light scattering occurs due to the phosphor inside when receiving the optical signal. This pc-white LED is not suitable for a receiving LED [24].

First, we measured the wavelength and optical power of the LEDs to find out the detailed characteristics of the transmitter and the receiver. The measured optical spectra of the LEDs are shown in Fig. 3. And we measured frequency response and signal-to-noise ratio (SNR) to find out the light-receiving characteristics of the traffic light. Figure 4 shows the measured optical power of the headlamp as a function of the

Table 1 Characteristics of headlamps and LED traffic lights

Parameter		Value	Unit
Peak Wavelength	Headlamp	560	nm
	Red	628	nm
	Yellow	593	nm
	Green	523	nm
Optical Power (When Applying 18 V)	Headlamp	29.6	mW

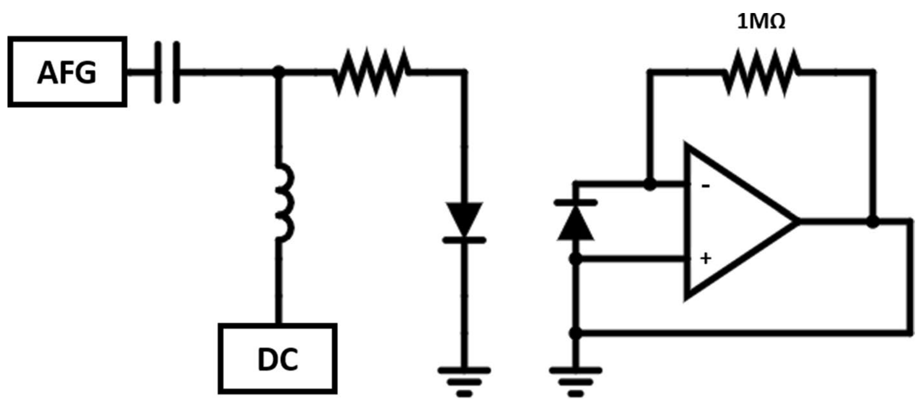

Fig. 5 Experimental setup of the headlamp-to-traffic light VLC

input voltage. When applying 18.0 V to the headlamp, the optical power of 29.6 mW was measured. The characteristics of the headlamps and traffic lights are summarized in Table 1.

We used a bias tee circuit at the transmitter. The direct current (DC) voltage supplied from a DC Power supply and the signal from an arbitrary function generator (AFG) were combined and inputted into the headlamp. In the receiver circuit, the signal was amplified with an OP Amp and measured by an oscilloscope. Zero bias was applied to the receiver circuit. Because the current generated by the receiving LED is lower than that of the PD, we used a high load resistance of 1 MΩ. The circuit for the vehicle-headlamp-to-traffic-light LED-to-LED VLC experiment is shown in Fig. 5.

To investigate the linear operation region of the LED-to-LED VLC, we measured the input–output function of the system, as shown in Fig. 6. The linear-operating region is also shown in Fig. 6. Linearity was confirmed when the transmitter voltages for red, yellow, and green receiver LEDs were 18–21 V, 17–19 V, and 17.5–20.5 V, respectively.

The frequency response and SNR were also measured by receiver LED color. Figure 7 shows the frequency response according to the color of the receiver. The 3-dB bandwidths with red, yellow, and green LED receivers were identified as 850, 855, and 809 Hz, respectively. In addition, in order to investigate the degree of noise of the signal, the received signal and noise were measured to check the SNR. SNRs in red, yellow, and green LED receivers were measured as 5.1 dB, 6.0 dB, and 6.4 dB, respectively. The 3-dB bandwidths and SNRs of the measured color-specific traffic lights are summarized in Table 2.

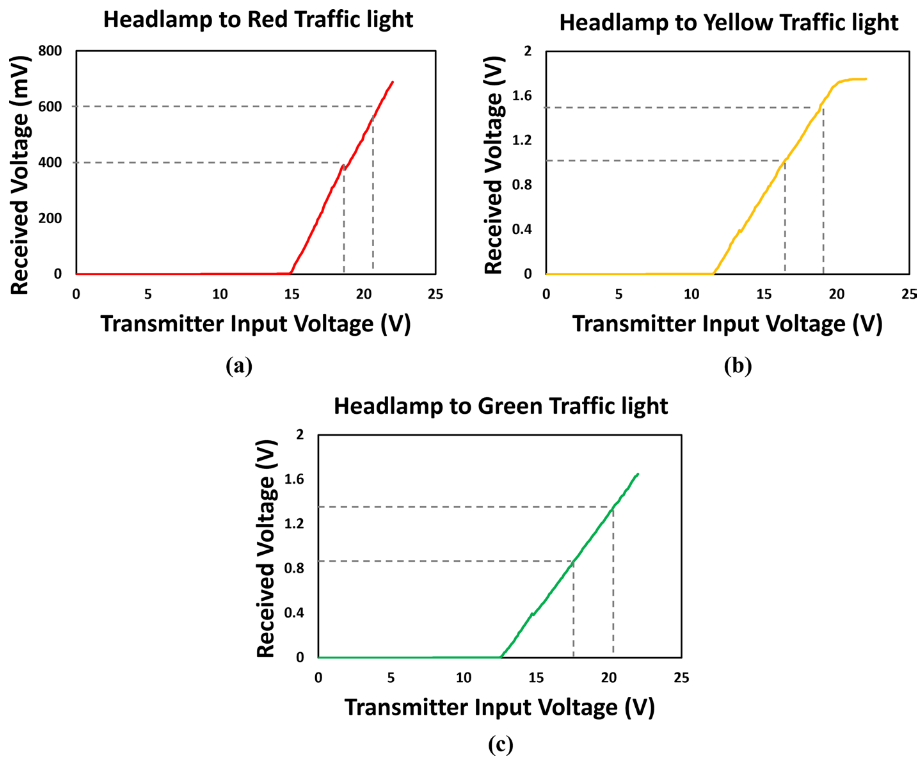

Fig. 6 Receiver voltage as a function of transmitter voltage in **a** headlamp-to-red LED, **b** headlamp-to-yellow LED, and **c** headlamp-to-green LED VLC system

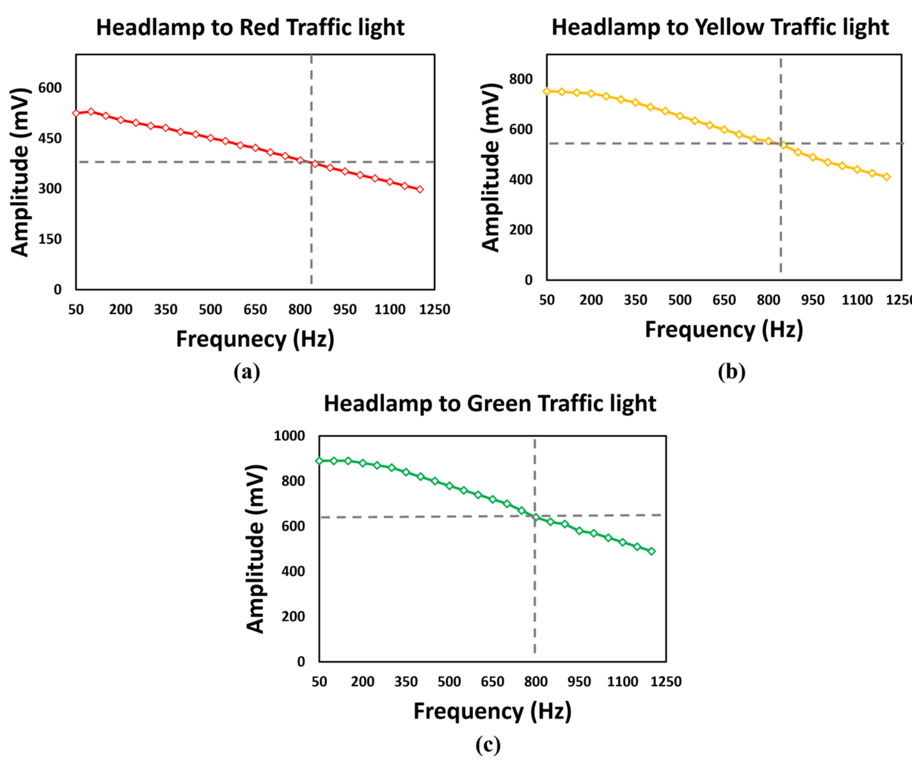

Fig. 7 Frequency response of **a** red, **b** yellow, and **c** green traffic lights

Table 2 Measured 3-dB bandwidth and SNR

Parameter		Value	Unit
3-dB Bandwidth	Red	850	Hz
	Yellow	855	Hz
	Green	809	Hz
SNR	Red	5.1	dB
	Yellow	6.0	dB
	Green	6.4	dB

To predict the theoretical channel capacity between the headlamp and the traffic light, Shannon's channel capacity equation was used [14]. Shannon's channel capacity equation is given by.

$$C \cong f_{3dB} * log_2(1 + SNR), \tag{1}$$

where C is the capacity of the channel, f_{3dB} is the 3-dB bandwidth of the channel, and SNR is the ratio of the average power of the signal to the average power of the noise, which is given by.

$$SNR = \frac{E\{\overline{p_s}\}}{E\{\overline{p_n}\}} \tag{2}$$

In Eq. (2), Ps is the power of the signal and Pn is the power of the noise. Then Shannon's channel capacity can be expressed using Eqs. (1) and (2),

$$C \cong f_{3dB} \cdot log_2\left(1 + \frac{E\{\overline{P_s}\}}{E\{\overline{P_n}\}}\right) \tag{3}$$

The channel capacity calculated by the derived Eq. (3) was 17.2, 17.1, and 14.5 kbps for headlamp-to-green, headlamp-to-yellow, and headlamp-to-red system, respectively.

After investigating the characteristics of LEDs, the communication experiment was conducted. VLC system is mainly based on intensity modulation direct detection (IM/DD) because it uses LEDs with incoherent properties. In the experiment, the orthogonal frequency division multiplexing (OFDM) technique was used. The reason why we chose OFDM technique was that it can efficiently use the frequency band due to orthogonal subcarriers and can minimize the effect of inter-symbol interference (ISI) by adding cyclic prefixes (CP) or zero padding (ZP) between symbols. However, the VLC system can only transmit unipolar real signals, making it difficult to use conventional OFDM modulation techniques. Therefore, the OFDM modulation techniques have been studied in various ways, including DC-biased optical OFDM (DCO-OFDM), asymmetrically clipped optical OFDM (ACO-OFDM), and asymmetrically clipped DC-biased optical OFDM (ADO-OFDM).

In this experiment, we used DCO-OFDM, which has higher spectral efficiency compared to other modulation techniques and relatively simple modulation and demodulation [14]. The block diagram of the DCO-OFDM modulation is shown in Fig. 8. The actual picture of the experiment setup is shown in Fig. 9.

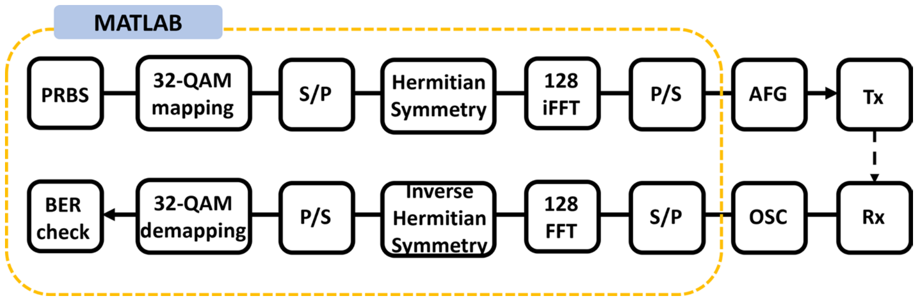

Fig. 8 Block diagram of DCO-OFDM modulation and demodulation

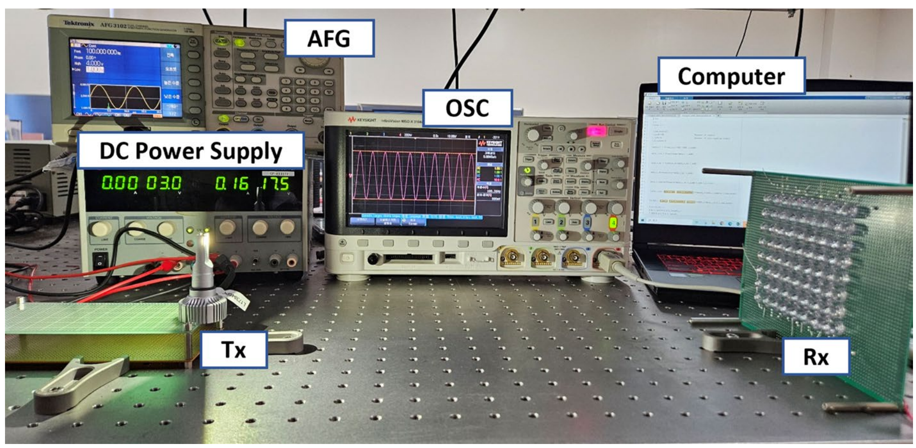

Fig. 9 Photo of the experimental setup for headlamp-to-traffic light VLC

3 Results and discussion

At the transmitter site, a pseudo random binary sequence (PRBS) signal was made through MATLAB, then it was modulated to a 32-quadrature amplitude modulation (QAM) signal. Next, for application to IM/DD systems, the Hermitian symmetry technique at the input of inverse fast Fourier transform (IFFT) was used to make the IFFT output a real value. A DC value was then added so that it could have unipolarity. Therefore, the signal that has a unipolar real value can be input to the headlamp LED, the transmitter, through AFG.

At the receiver site, the signal received at the traffic light LED was demodulated in reverse to the process of modulation. The demodulated signal was measured through an oscilloscope and the bit-error rate (BER) was checked by MATLAB software. In general, a BER threshold of 10^{-3} is used in wireless communication, taking the forward error correction (FEC) code into account. Therefore, in this study, the BER threshold for reliable communication was also assumed to be 10^{-3}. Additionally, the signal was measured based on the data rate. The graph of the BER representing the communication performance is shown in Fig. 10, and the scatterplot is shown in Fig. 11. It was confirmed that the headlamp-to-traffic-light LED-to-LED VLC system can communicate at a data rate of 5.8 kbps for green and yellow LED receiver, and 5.1 kbps for red LED receiver. This indicates that it is possible for the vehicle's headlamp to communicate with traffic lights.

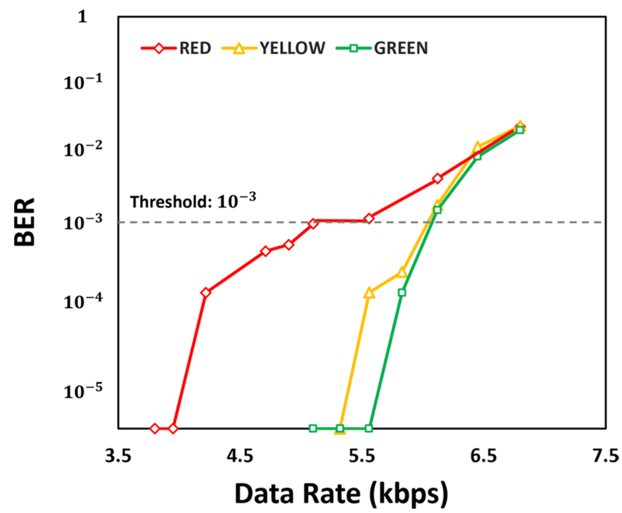

Fig. 10 BER of the headlamp-LED-to-traffic-light-LED system

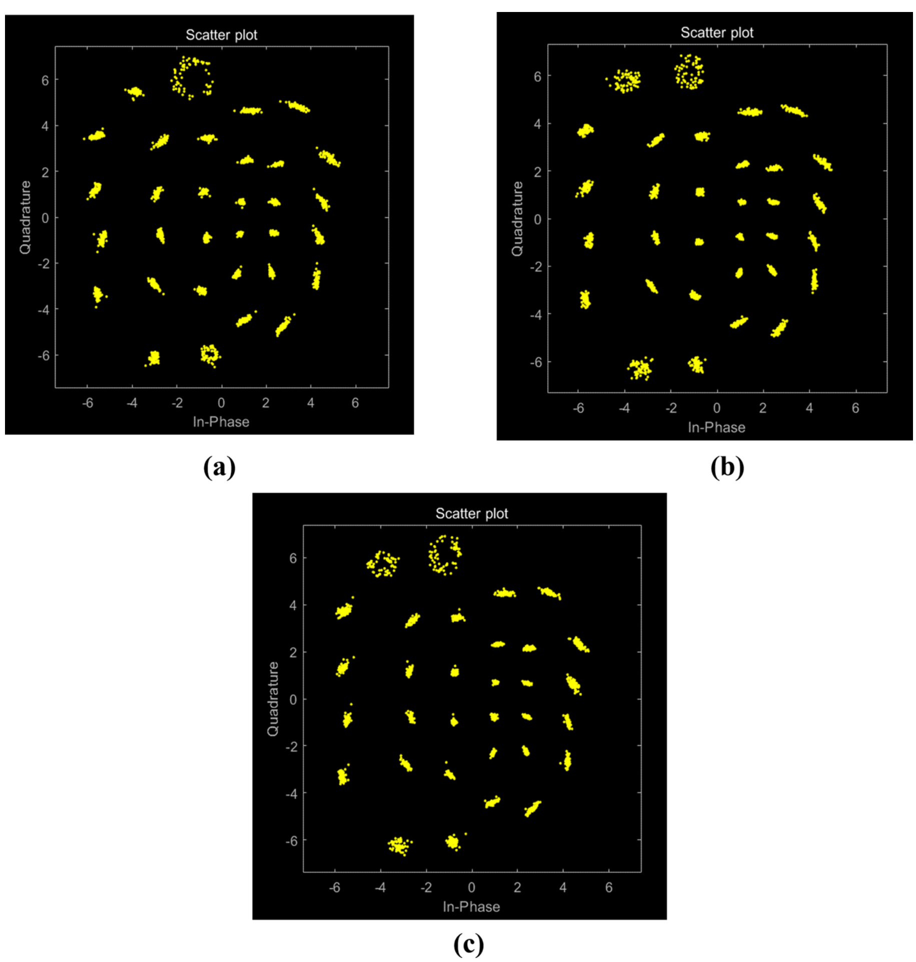

Fig. 11 Scatterplot of **a** headlamp-to-red-traffic-light, **b** headlamp-to- yellow-traffic-light, and **c** headlamp-to-green-traffic-light

4 Conclusion

In this work, we demonstrate a LED-to-LED VLC using a vehicle headlamp and a traffic light. Red, yellow, and green traffic light LEDs are used in the experiment. To eliminate distortion of the signal, the linear operation region was checked, and the frequency response and SNR of the LED-to-LED VLC were measured. Based on these results, the channel capacity was analyzed using Shannon's channel capacity formula. Finally, the experiment was conducted using the 32-QAM DCO OFDM modulation technique. As a result, it was confirmed that the green and yellow traffic light LED receivers were capable of 5.8 kbps of communication, and the red traffic light LED receivers were capable of 5.1 kbps of communication.

Here, we demonstrated that the LED-to-LED VLC can be applied to the field of vehicle communication. In this study, LED-to-LED VLC technology is expected to be utilized in the actual vehicle communication field in the long term to contribute to solving the RF band supersaturation problem. Unlike conventional visible light communication, it does not require additional optical elements, which can provide economic benefits. We think that this technique can be used for data communication on the road, such as vehicle data collection, traffic light control, and road traffic conditions. In addition, the technology has the potential to develop not only vehicle communication but also other fields such as aviation, indoor communication, and IoT.

This study is an experimental investigation into the potential use of the LED-to-LED VLC system for vehicle communication. However, since the experiments were not conducted in real-world environments, additional research is needed to address the effects of surrounding factors such as shadows, lighting, and obstacles. The areas for improvement in this experiment are the threshold value of BER, short transmission distance, and the fact that the communication system was designed based on the assumption that the vehicle is stationary. In road environments, high reliability and performance are required, so a BER threshold of at least 10^{-5} should be applied. To address this issue, MIMO technology, which can reduce BER, is expected to be a viable solution. Therefore, further research on the LED-to-LED VLC system is needed, focusing on the BER threshold, extending the transmission distance, and vehicle mobility. Research on the LED-to-LED VLC system is currently ongoing, and it is expected that the challenges encountered will be addressed in the future.

Abbreviations

VLC	Visible light communication
LED	Light-emitting diode
RF	Radio frequency
SNR	Signal-to-noise ratio
BER	Bit-error rate
OWC	Optical wireless communication
IR	Infrared
UV	Ultraviolet
VL	Visible light
Li-Fi	Light fidelity
OCC	Optical camera communication
FSOC	Free space optical communication
LD	Laser diode
PD	Photodiode
IS	Image sensor
IoT	Internet of things
EMI	No electromagnetic interference

MIMO	Multiple-input multiple-output
ADAS	Advances driver assistance systems
DSRC	Dedicated short-range communication
I2V	Infrastructure-to-vehicle
V2V	Vehicle-to-vehicle
PPM	Pulse position modulation
pc-white LED	Phosphorescent conversion white LED
DC	Direct current
AFG	Arbitrary function generator
IM/DD	Intensity modulation direct detection
OFDM	Orthogonal frequency division multiplexing
ISI	Inter-symbol interference
CP	Cyclic prefixes
ZP	Zero padding
DCO-OFDM	DC-biased optical OFDM
ACO-OFDM	Asymmetrically clipped optical OFDM
ADO-OFDM	Asymmetrically clipped DC-biased optical OFDM
PRBS	Pseudo random binary sequence
QAM	Quadrature amplitude modulation
IFFT	Inverse fast Fourier transform
FEC	Forward error correction

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Author contributions
Se-Jin Park conducted experiments, did the data analysis, and wrote the first draft of manuscript; Sung-Man Kim proposed the main idea, designed the work, and revised the manuscript.

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Data availability
Data generated or analyzed during this study are included in this published article.

Declarations

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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